

# Writing Large Numbers in Scientific Notation

*Teacher's Edition*

## The Period

**MA.8.NSO.1.4** Express numbers in scientific notation to represent and approximate very large or very small quantities. Determine how many times larger or smaller one number is compared to a second number.

**Must** Express very large quantities in scientific notation and read them back into standard form; determine how many times larger one number is than another.

**Must not** Coefficients are at least 1 and less than 10. Large numbers today; negative powers of ten are 3.09, and operations in scientific notation are Unit 4.

**Target** I can write a large number in scientific notation, read one back into standard form, and find how many times larger one number is than another.

**MTR** **MTR.7.1** — Every board and the bank · **MTR.5.1** — Notes I · **MTR.6.1** — Board 4

| Segment                             | Slides | Time          |
|-------------------------------------|--------|---------------|
| Settle in; post the learning target | 1      | 1 min         |
| Warm-Up                             | 2-3    | 5 min         |
| Notes                               | 4-6    | 11 min        |
| Example 1                           | 7-8    | 3 min         |
| Your Turn                           | 9-10   | 3 min         |
| Example 2                           | 11-12  | 3 min         |
| Your Turn                           | 13-14  | 3 min         |
| Whiteboards                         | 15-32  | 13 min        |
| Independent Set                     | 33     | 6 min         |
| IXL                                 | 34     | 5 min         |
| <b>Total</b>                        |        | <b>53 min</b> |

### Say these three things out loud today

1. Exactly one nonzero digit sits in front of the decimal point. One — not two, and not none.
2. The exponent counts the places the decimal point moved. It does not count zeros.
3. To compare, read the exponent first. Only when the exponents match does the coefficient decide.

## Slide by Slide

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1. **Writing Large Numbers in Scientific Notation.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes..** Q2 and Q4 are the two halves of today: a power of 10 names a place.
3. **Warm-Up — Answers..** Reveal. 1.  $4/x^3$  2.  $10^6 = 1,000,000$  3. 11 4. millions
4. **Notes — Copy the table..** The exponent is the number of zeros, and it is also the place of the 1:  $10^3$  is one thousand, and the 1 sits in the thousands place.
5. **Notes.** One nonzero digit in front of the decimal point. That is the whole rule.
6. **Notes.**  $10^3$  times larger is a thousand times larger. The exponent is not the answer; the power is.
7. **Example 1 — Standard form to scientific notation..** SPOKEN ONLY.
8. **Example 1 — Worked..** The 3 is in the ten-billions place:  $10^{10}$ .
9. **Your Turn — Same steps, your numbers. Boards up when done..** The book's auction painting.
10. **Your Turn — Answer..** Reveal; name what a wrong board most likely did (see the note above).
11. **Example 2 — How many times larger?.** SPOKEN ONLY.
12. **Example 2 — Worked..**  $8.1 \div 5.4 = 1.5$  and  $10^8 \div 10^7 = 10$ .
13. **Your Turn — Same steps, your numbers. Boards up when done..** The book's alligators.
14. **Your Turn — Answer..** Reveal; name what a wrong board most likely did (see the note above).
- 15-16. **Board 1.** Write 4,000,000 in scientific notation. →  **$4 \times 10^6$**   $4 \times 10^7$  — counted the digits instead of the places the point moves [B1G-M 8.NSO.1.4: exponent = places moved];  $40 \times 10^5$  — coefficient not in [1, 10] [B1G-M 8.NSO.1.4 misconception: coefficient outside 1 to 10]
- 17-18. **Board 2.** Write  $7.2 \times 10^4$  in standard form. → **72,000** 720,000 — the point was moved five places [8.NSO.1.4:  $10^4$  moves the point four places]; 7.20000 — zeros appended without moving the point
- 19-20. **Board 3.** A stadium's renovation used 356,000 bolts. The stadium seats 71,000 people. Write the number of bolts in scientific notation. →  **$3.56 \times 10^5$**  The seating figure is not needed.  $356 \times 10^3$  — coefficient not in [1, 10] [B1G-M 8.NSO.1.4 misconception];  $3.56 \times 10^3$  — counted the zeros instead of the places the point moves;  $7.1 \times 10^4$  — the wrong number was converted
- 21-22. **Board 4.** Which is greater,  $9.5 \times 10^5$  or  $1.2 \times 10^6$  ? →  **$1.2 \times 10^6$**   $9.5 \times 10^5$  — compared the coefficients and ignored the powers [B1G-M 8.NSO.1.4: order of magnitude first]
- 23-24. **Board 5.** Which is 52,000,000 written in scientific notation? →  **$A - 5.2 \times 10^7$**  (B) equal in value, but the coefficient 52 is not less than 10 [B1G-M 8.NSO.1.4 misconception: coefficient outside 1 to 10]; (C) counted the six zeros instead of the seven places the point moves [8.NSO.1.4: exponent = places moved]; (D) counted the eight digits [8.NSO.1.4: exponent = places moved] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.
- 25-26. **Board 6.** How many times larger is  $6 \times 10^8$  than  $6 \times 10^5$  ? →  **$A - 1,000$**  (B)  $10^3$  times read as 3 times [B1G-M 8.NSO.1.4 misconception: ' $10^4$  times' read as

'4 times']; (C)  $3 \times 100$  — the exponent difference multiplied by a stray 100 [8.NSO.1.4: the ratio of the powers is  $10^3$ ]; (D)  $8 \div 5$  — the exponents were divided [8.NSO.1.3: quotient of powers subtracts the exponents] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.

**27-28. Board 7.** How many times larger is  $8.4 \times 10^6$  than  $2.1 \times 10^4$  ? → **400**  
4 — coefficients only [8.NSO.1.4: both factors are compared]; 40 —  $10^2$  read as 10 [B1G-M 8.NSO.1.4 misconception: exponent read as the multiplier]

**29-30. Board 8.** A computer displays a number as 3.5E9. Write the number in standard form. → **3,500,000,000** 'error' or  $3.5^9$  — E read as something other than  $\times 10^9$  [B1G-M 8.NSO.1.4 misconception: calculator E notation]; 35,000,000,000 — point moved ten places

**31-32. Board 9.** Marcus writes 2,000,000 as  $20 \times 10^5$ . Is his expression equal to 2,000,000? Is it scientific notation? Explain. → **Equal, yes. Scientific notation, no: 20 is not less than 10.  $2 \times 10^6$ .** 'not equal' — value confused with form [8.NSO.1.4: equivalent expressions can differ in form]; 'yes, it is scientific notation' — the coefficient constraint was not applied [B1G-M 8.NSO.1.4 misconception]

**33. Independent Set — Six questions. On your own, in silence..** Hand out the Independent Set.

**34. IXL — Last five minutes..** IXL: every listed skill is required, SmartScore 67, due at the start of the next class.

## Misconceptions to Watch

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**Board 1.**  $4 \times 10^7$  — counted the digits instead of the places the point moves [B1G-M 8.NSO.1.4: exponent = places moved]

**Board 2.** 720,000 — the point was moved five places [8.NSO.1.4:  $10^4$  moves the point four places]

**Board 3.**  $356 \times 10^3$  — coefficient not in [1, 10] [B1G-M 8.NSO.1.4 misconception]

**Board 4.**  $9.5 \times 10^5$  — compared the coefficients and ignored the powers [B1G-M 8.NSO.1.4: order of magnitude first]

**Board 5.** equal in value, but the coefficient 52 is not less than 10 [B1G-M 8.NSO.1.4 misconception: coefficient outside 1 to 10]

**Board 6.**  $10^3$  times read as 3 times [B1G-M 8.NSO.1.4 misconception: ' $10^4$  times' read as '4 times']

**Board 7.** 4 — coefficients only [8.NSO.1.4: both factors are compared]

**Board 8.** 'error' or  $3.5^9$  — E read as something other than  $\times 10^9$  [B1G-M 8.NSO.1.4 misconception: calculator E notation]

**Board 9.** 'not equal' — value confused with form [8.NSO.1.4: equivalent expressions can differ in form]

**Materials** Whiteboards and markers. Calculators are allowed on everything; today is the day to show the class what E means on the display — enter  $3.5 \times 10^9$  and 35,000,000,000 and look.